

Chapter 2 Exploring Data with Tables and Graphs Scatterplots, Correlation, and Regression (2.4)

Objectives

In this lesson, we will learn how to

- analyze paired sample data;
- use scatterplots to look for relationships between two quantitative variables;
- understand the meaning of correlation and linear correlation;
- interpret the linear correlation coefficient r and a P-value;
- describe the idea of a regression line.

1. Paired Data

Many statistical questions involve two measurements from the same individual or object.

Paired Data

Paired data occur when each observation has two related quantitative values, usually written as

$$(x, y).$$

For example, one person may have both a shoe print length and a height.

When data are paired, we are often interested in whether one variable is associated with the other.

2. Scatterplots

Scatterplot

A scatterplot is a graph of paired (x, y) quantitative data.

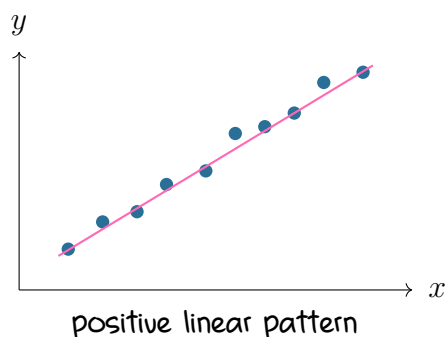
- The horizontal axis is used for the first variable, x .
- The vertical axis is used for the second variable, y .
- Each ordered pair (x, y) is plotted as a point.

Example 1: A researcher records the overhead width and weight of several seals. What would a clear upward pattern in the scatterplot suggest?

Solution

A clear upward pattern suggests that the two variables are associated. In this setting, seals with larger overhead widths tend to have larger weights.

This indicates a positive relationship between overhead width and weight.



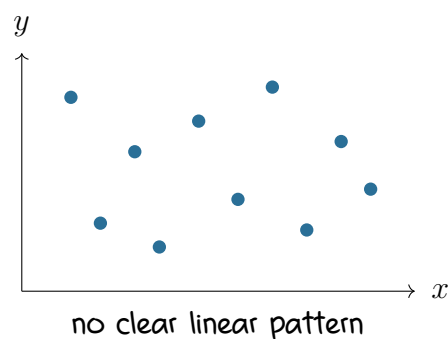
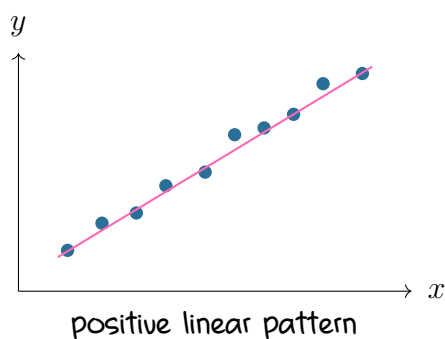
3. Correlation and Linear Correlation

Correlation

A **correlation** exists between two variables when the values of one variable are somehow associated with the values of the other variable.

Linear Correlation

A **linear correlation** exists when the plotted points show a pattern that can be reasonably approximated by a straight line.



Example 2: Suppose a scatterplot of presidents' heights and their main opponents' heights shows no clear pattern. What conclusion should we make?

Solution

If the plotted points show no clear pattern, then there does not appear to be a correlation between the two variables.
In particular, there does not appear to be a linear correlation.

4. Linear Correlation Coefficient

Linear Correlation Coefficient

The linear correlation coefficient, denoted by r , measures the strength and direction of the linear association between two quantitative variables.

Interpreting r

$$-1 \leq r \leq 1.$$

- If r is close to 1, there is a strong positive linear correlation.
- If r is close to -1 , there is a strong negative linear correlation.
- If r is close to 0, there is little or no linear correlation.

Value of r	Meaning
$r \approx 1$	Strong positive linear relationship
$r \approx -1$	Strong negative linear relationship
$r \approx 0$	No strong linear relationship

5. Example: Shoe Print Lengths and Heights

Example 3: The following paired data give shoe print length and height for five people.

Shoe Print Length (cm)	29.7	29.7	31.4	31.8	27.6
Height (cm)	175.3	177.8	185.4	175.3	172.7

Suppose technology gives

$$r = 0.591, \quad P\text{-value} = 0.294.$$

Is there sufficient evidence of a linear correlation?

Solution

The value $r = 0.591$ suggests a moderate positive linear relationship, but the sample size is very small.

The P-value is

$$0.294.$$

This is not small. Since $0.294 > 0.05$, the result could easily occur by chance if there were no true linear correlation.

Therefore, there is not sufficient evidence to conclude that shoe print length and height have a linear correlation based on these five pairs of data.

6. P-Values for Correlation

P-Value for Linear Correlation

If there really is no linear correlation between two variables, the P-value is the probability of getting paired sample data with a value of r at least as extreme as the one obtained from the sample.

How to Interpret the P-Value

- A small P-value, such as 0.05 or less, supports a conclusion that there is a linear correlation.
- A large P-value does not provide enough evidence to conclude that there is a linear correlation.

Example 4: For a larger data set with $n = 40$, suppose technology gives

$$r = 0.813, \quad P\text{-value} = 0.000.$$

What conclusion should we make?

Solution

The value $r = 0.813$ is close to 1, so the sample shows a strong positive linear relationship.

The P-value is very small. Therefore, we have sufficient evidence to conclude that there is a linear correlation between shoe print lengths and heights.

7. Regression

Regression Line

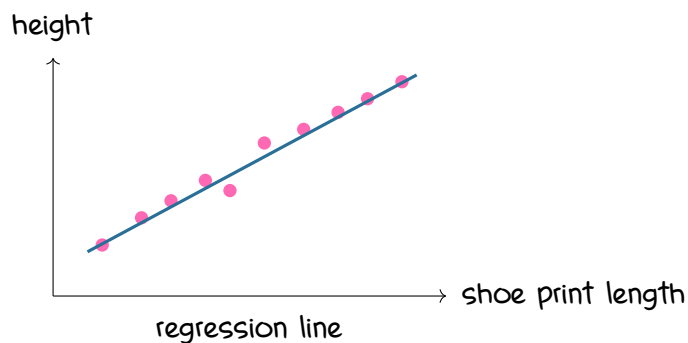
Given paired sample data, the regression line, also called the line of best fit or least-squares line, is the straight line that best fits the scatterplot of the data.

The general form of the regression equation is

$$\hat{y} = b_0 + b_1x,$$

where

- b_0 is the y -intercept;
- b_1 is the slope;
- $\hat{y}$ is the predicted value of y .



Example 5: Suppose the regression equation for predicting height from shoe print length is

$$\text{Height} = 80.9 + 3.22(\text{Shoe Print Length}).$$

Use the equation to predict the height of a person with a shoe print length of 30 cm.

Solution

Substitute 30 for shoe print length:

$$\text{Height} = 80.9 + 3.22(30).$$

$$= 80.9 + 96.6$$

$$= 177.5.$$

The predicted height is

$$177.5 \text{ cm}.$$

8. Important Cautions

Correlation Does Not Mean Causation

Even if two variables have a strong correlation, this does not prove that one variable causes the other.

Use Regression Carefully

- Use a scatterplot before relying on a regression equation.
- Do not use a regression equation for predictions far outside the range of the original data.
- Remember that predictions are estimates, not exact values.

Summary

Key Ideas

- A scatterplot displays paired quantitative data.
- Correlation means that two variables are associated.
- Linear correlation means the pattern can be approximated by a

straight line.

- The correlation coefficient r is always between -1 and 1 .
- A small P-value supports the conclusion that linear correlation exists.
- A regression line is used to predict y from x .
- Correlation alone does not prove causation.